$$X(f) = 2V_0 T \operatorname{sinc}(2fT) e^{-j2\pi fT} \frac{1}{4T} \sum_k \delta\left(f - \frac{k}{4T}\right)$$

$$= \frac{V_0}{2} \operatorname{sinc}(2fT) e^{-j2\pi fT} \sum_k \delta\left(f - \frac{k}{4T}\right)$$

$$= \frac{1}{4T} \sum X_p(f) \, \delta\left(f - \frac{k}{4T}\right)$$

$$= \sum \frac{1}{4T} X_p\left(\frac{k}{4T}\right) \delta\left(f - \frac{k}{4T}\right)$$

$$d_k = \frac{X_p\left(\frac{k}{4T}\right)}{4T}$$

calcolo di α_k

$$\alpha_k = \frac{1}{4T} X_p\left(k\frac{1}{4T}\right)$$

$$= \frac{1}{4T} \cdot 2V_0 T \operatorname{sinc}\left(2(\pi)T\frac{k}{4T}\right) e^{-j2\pi T\frac{k}{4T}}$$

$$\alpha_0 = \frac{1}{4T} 2V_0 T \cdot 1 \cdot 1 = \frac{V_0}{2}$$

$$\alpha_1 = \frac{1}{4T} 2V_0 T \operatorname{sinc}\left(\frac{2(\pi)T\,1}{4T}\right) e^{-j2\pi T\frac{1}{4T}}$$

$$= \frac{1}{4T} 2V_0 T \operatorname{sinc}\left(\frac{(\pi)}{2}\right) e^{-j\frac{\pi}{2}}$$

$$= -\frac{V_0}{\pi} j$$

$$\alpha_{-1} = +\frac{V_0}{\pi} S$$

$$\alpha_{2k} = 0 \quad \forall k$$

N.B.: $\text{sinc}\,\frac{1}{2}$ si intende $\text{sinc}\,\frac{\pi}{2}$